Equations

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May 2025

1 Global Economic Growth

Let us first establish our variables for determining Global Economic Growth.

1.1 Variables

- WN is the set of all nations in the world.
- c_{GDP-I} is the initial GDP of country c. The initial GDP of country c is country c's GDP at the beginning of the game.
- c_{GDP} is the current GDP of country c. The current GDP of country c is country c's GDP after additional tarriffs have been applied.
- c_T is the tariff rate on country c.
- c_X is the value of exports from country c to the US.
- ϵ is export price elaciticity. ϵ is usually between -0.5 and -1.5, so we will assume $\epsilon = -1$ in all cases and for all countries.
- $\triangle X$ is the percent change in exports to the US. $\triangle X = -100\%$ wipes out all exports, so no further changes should happen with additional tariffs after this point. Maybe having a tariff rate of 100% could unlock a prize or special feature. This also means that c_T cannot actually go over 100%, so the clicker should stop working after that point.
- $\triangle P$ is the percent increase in effective price due to tariff. For the sake of simplicity, we will assume that $\triangle P$ is equal to the US tariff rate on a specific country. Therefore, $c_{\triangle P} = c_T$.

Now let us establish the equations that we will use to translate tariff rates across different countries to determine Global Economic Growth.

1.2 Equations

- Initial GDP $= \sum_{c \in WN} c_{GDP-I}$. Initial Global GDP is the sum across all countries in the world of the initial GDP of each country.
- Current Global GDP $= \sum_{c \in WN} c_{GDP}$. Current Global GDP is the sum across all countries in the word of the current GDP of each country.
- Basic elasticity formula is $c_{\triangle X} = \epsilon \cdot c_{\triangle P}$. Since $\epsilon = -1$ and $c_{\triangle P} = c_T$, then $c_{\triangle X} = c_T$.
- $c_{GDP} = c_{GDP-I} - 1.5 \cdot c_T \cdot c_X$. The 1.5 is an export multiplier that empirical studies have identified to assess the effects of trade losses due to tariffs on GDP. The value is different for different countries, but we are assuming a value of 1.5 for all countries.
- (Current Global GDP - Initial Global GDP)/Initial GDP = Global Economic growth.

1.3 Notes

- IMF estimates from past tariff waves suggest each $\$100B$ in global trade loss reduces global GDP by 0.1–0.2% [median value in this range 0.15]
- $\$468.8B$ loss $\cdot 0.15\%$ per $\$100B = ~0.7\%$ global GDP decline

2 Net US Jobs

2.1 Variables

- $\triangle M$ is the reduction in US imports due to US tariffs.
- α is the share of reduction that is replaced by US production. In theory, α is greater than 0 and less than 1. However, in the US α typically ranges between 0.2 and 0.6. Perhaps, we could have different α values for easy and hard modes, or higher α value could be something that a player can unlock.
- β is # of jobs per $\$1B$ of output in the protected sector. Typically, β is between 2000 and 10000.
- R is the total tariff revenue.
- γ is the downstream job multiplier, expressed as # of jobs per $\$1B$ of loss. γ is typically between 3000 and 6000.

2.2 Equations

- $\triangle M = \sum_{c \in WN}(c_T \cdot c_X)$.
- $R = \triangle M/\epsilon$, where ϵ represents price elasticity. As we mentioned in section 1.1, we are assuming that $\epsilon = -1$ in all cases, so we can use the simpler equation $R = -\triangle M$.
- US Jobs Gained in Protected Sectors $= \triangle M \cdot \alpha \cdot \beta$.
- US Jobs Lost in Downstream Sectors $= -R \cdot \gamma = \triangle M \cdot \gamma$.
- Net US Jobs = US Jobs Gained in Protected Sectors - US Jobs Lost in Downstream Sectors.

2.3 Notes

- If we were to factor in retaliatory tariffs, the equation for net US jobs would be Net US Jobs = Jobs Gained in Protected Sectors - Jobs Lost in Downstream Sectors - Jobs Lost from Retaliation, where Jobs Lost from Retaliation = (U.S. exports affected by retaliation) $\cdot$ (export job multiplier (expressed in terms of jobs per \$1 billion exports)).

3 Consumer Prices

3.1 Variables

- The Pass Through Rate measures how much of the tariff is passed to consumers. Studies have found pass-through rates from 50% to over 100%, depending on market power, competition, and product type. For the sake of simplicity, let us assume a universal pass through rate, denoted as P.
- $\triangle P$ is the % change is consumer prices.

3.2 Equations

- $\triangle P = \sum_{c \in WN} c_T \cdot c_X \cdot P$

3.3 Notes

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4 Trade Volume

4.1 Variables

- c_{XW-I} is the initial total value of all exports from country c. The initial total value of all exports from country c is the total value from of all exports from country c at the beginning of the game.

- c_{XW} is the current total value of all exports from country c. The current total value of all exports from country c is the total value from of all exports from country c after additional tariffs.

4.2 Equations

- $c_{XW} = c_{XW-I} - 0.6 \cdot c_T \cdot c_X$. The 0.6 is included due to trade diversion.
- Initial Global Trade Volume $= \sum_{c \in WN} c_{XW-I}$. Initial Global Trade Volume is the sum across all countries in the world of the initial total value of all exports for each country.
- Current Global Trade Volume $= \sum_{c \in WN} c_{XW}$. Current Global Trade Volume is the sum across all countries in the world of the current total value of all exports for each country.
- (Global Trade Volume - Initial Global Trade Volume)/Initial Global Trade Volume = change in trade volume.

4.3 Notes

- The research I performed and reviewed on the effect of tariffs on global trade volume included retaliatory tariffs from other countries, which exacerbate losses further. For simplicity of coding and ease of game play, I decided not to include retaliatory tariffs in my equations.